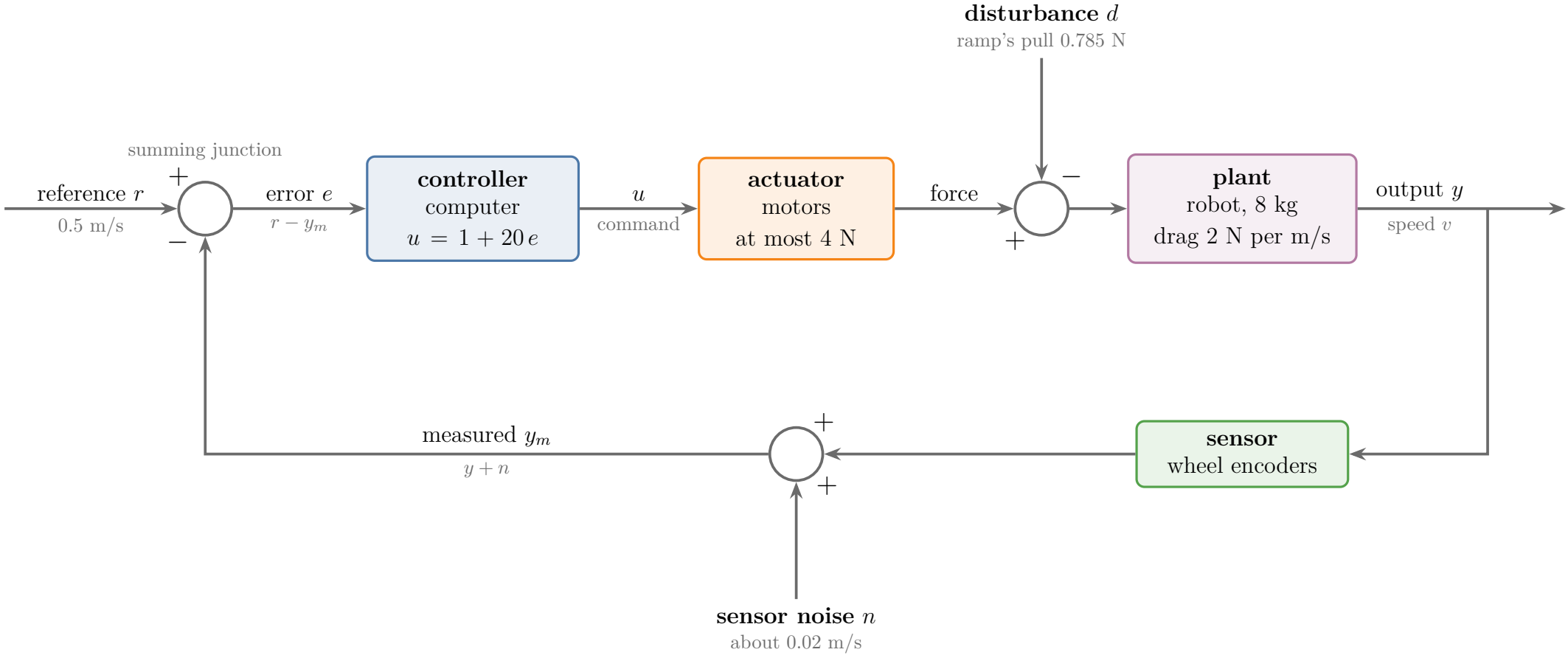

disturbance d
ramp's pull 0.785 N
summing junction
reference r
0.5 m/s
+
−
error e
$r - y_m$
controller
computer
$u = 1 + 20\,e$
u
command
actuator
motors
at most 4 N
force
−
+
plant
robot, 8 kg
drag 2 N per m/s
output y
speed v
measured y_m
$y + n$
+
+
sensor
wheel encoders
sensor noise n
about 0.02 m/s